Shri Vishnu Engineering College for Women

EMBEDDED SYSTEMS

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CSE-A

COMMUNICATION INTERFACE

Communication Interface is a method used for transferring data between

- Processor
- Memory
- Sensors
- Actuators
- External devices

It enables reliable communication between hardware components.

WHY COMMUNICATION INTERFACE IS NEEDED?

- Data transfer
- Device synchronization
- High speed communication
- Low power communication
- Reliable communication

Types of Communication Interface

1. Onboard Communication

Communication inside the embedded system.

Examples

- I^2C
- SPI
- UART
- CAN

2. External Communication

Communication with outside devices.

Examples

- USB
- Bluetooth
- Wi-Fi
- GSM
- ZigBee

I²C (INTER-INTEGRATED CIRCUIT)

- I²C (Inter-Integrated Circuit) is a communication system used to connect devices in an embedded system.
- It uses only two wires to transfer data between devices.
- It is a synchronous (clock-based), bi-directional (both devices can send and receive data), and half-duplex (data flows in only one direction at a time) communication bus.
- It was developed by Philips Semiconductors in the early 1980s.
- The main purpose of I²C was to provide an easy and efficient method of communication between a microprocessor/microcontroller and its peripheral devices, especially in television (TV) systems.

I²C Bus Lines

SCL (Serial Clock Line)

- SCL (Serial Clock Line) is used by the master device to generate the clock signal.
- The clock signal synchronizes the communication between the master and slave devices.
- It ensures that both devices send and receive data at the same speed.

SDA (Serial Data Line)

- SDA (Serial Data Line) is used to transmit and receive the actual data between devices.
- It is a bi-directional communication line, allowing both the master and slave devices to send and receive data.
- Data is transferred one bit at a time and is synchronized by the SCL clock signal.

SERIAL PERIPHERAL INTERFACE (SPI) BUS

Serial Peripheral Interface (SPI) Bus:

- SPI is a synchronous, bi-directional, full-duplex, 4-wire serial communication bus.
- Developed by Motorola.
- Commonly used to send data between microcontrollers and small peripherals like sensors, SD cards, shift registers, etc.

Basic Features:

- SPI follows a single-master, multi-slave system.
- More than one master can exist, but only one master should be active at a time.

SPI SIGNAL LINES

SPI uses four signal lines for data communication:

1. MOSI (Master Out, Slave In):

- Carries data from Master to Slave.

2. MISO (Master In, Slave Out):

- Carries data from Slave to Master.

3. SCLK (Serial Clock):

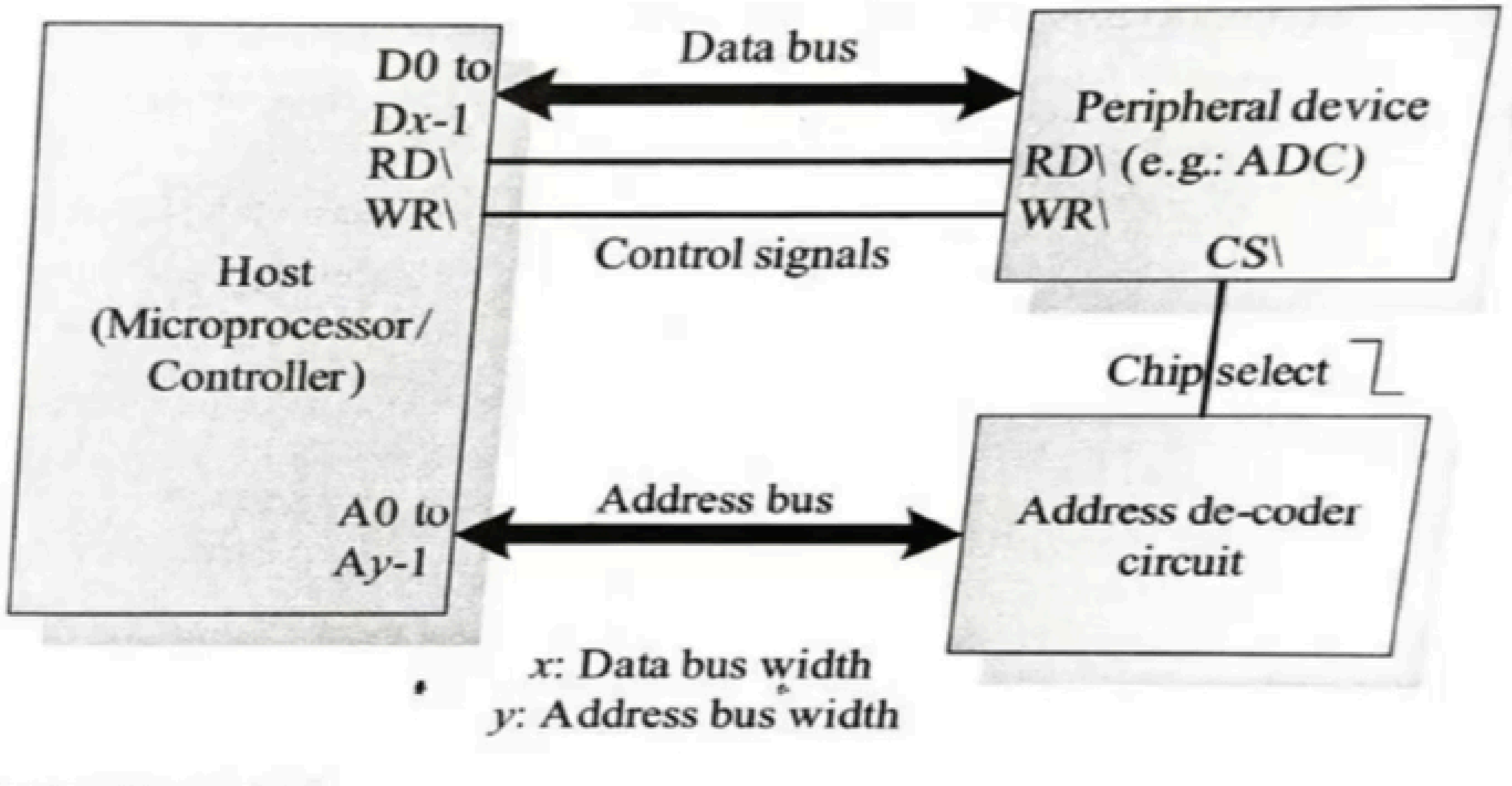
- Provides clock signals. Generated by Master.

4. SS (Slave Select):

- Used by Master to select which slave to communicate with.
- Active LOW signal.

Parallel Interface

- A parallel interface is used to connect the main controller (or processor) of an embedded system to other parts like displays, sensors, or memory.
- It sends many bits at once using multiple wires (unlike serial communication which sends one bit at a time).



Wired Communication Interfaces

A Wired Communication Interface is a way of sending data between two or more devices using physical wires or cables.

Types of Wireless Communication

1. USB

2. Ethernet

USB (Universal Serial Bus)

USB

- Universal Serial Bus
- High-speed serial communication
- Plug and Play
- Hot Swappable

Applications

- Keyboard
- Mouse
- Pendrive
- Printer
- Camera

Wireless Communication Interfaces

Wireless communication interfaces enable an embedded system to exchange data with other devices without using physical cables. They use radio waves, infrared signals, or cellular networks for communication.

Types of Wireless Communication

1. Infrared (IR)

2. Bluetooth

3. Wi-Fi

1. Infrared (IR)

- Uses infrared light for communication.
- Short-range and requires line-of-sight.
- Used in TV remotes and air conditioner remotes.

2. Bluetooth

- Short-range wireless communication (about 10 m).
- Low power consumption.
- Used in wireless headphones, keyboards, mice, and smartwatches.

3. Wi-Fi

- High-speed wireless communication.
- Provides Internet connectivity through a Wireless Local Area Network (WLAN).
- Used in laptops, smartphones, smart TVs, and IoT devices

THANK YOU